

IEOR E4004

Optimization Models and Methods

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Outline

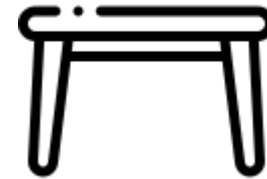
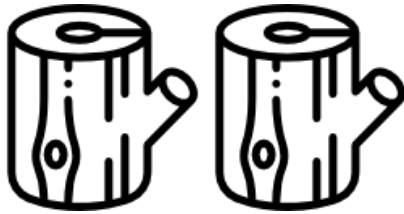
- Introduction to Mathematical Modeling
- Optimization Terminology
- Classes of Mathematical Programs

Introduction to Mathematical Modeling

A Production Problem

Mathematical Modeling

Consider a company produces chairs and tables:



Mathematical Modeling



Consider a company produces chairs and tables:

	Table	Chair
Wood need per unit	2	1
Hours of labor needs per unit	1	1
Profit of selling one item	\$3	\$2
Limit of selling each week	40	No limit

Each week, we can have **80 hrs of labor** and **100 units of wood** to spend.

Let us find out, what is the weekly production plan that maximize the profit?

A Production Problem: Formulation



The approach of optimization modeling involves developing a mathematical model by:

1. Describing which potential **decisions** could be taken,
2. Considering physical, logical, financial, or other **limitations**, and
3. Assessing them using a single **objective**

A Production Problem: Formulation



Guiding Questions:

What are the **potential decisions** that we can take?

What are our **constraints/ limitations**?

What is our **objective** in this problem?

A Production Problem: Formulation



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Decision Variables

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Objective

A Production Problem: Formulation



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Constraints

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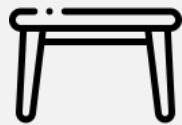
Let us find out, what is the weekly production plan that maximize the profit?

A Production Problem: Formulation



Step 1: Define decision variables

01



= Number of tables we should produce in a week



= Number of chairs we should produce in a week

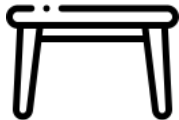
A Production Problem: Formulation

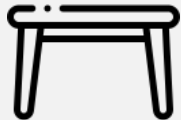


Step 2: Form objective functions

Q: What is our aim/objective?

→ To find the production plan that will **maximize** the profit

Profit: 3  + 2 

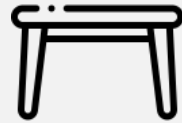
02 Aim/objective: **maximize** 3  + 2 

A Production Problem: Formulation



Step 3: Add constraints

Remember:



produced each week = x_1



produced each week = x_2

A Production Problem: Formulation



Step 3: Add constraints

03

subject to: 2  +  ≤ 100 Capacity of wood

A Production Problem: Formulation



Step 3: Add constraints

subject to $2 \begin{array}{c} \text{---} \\ | \quad | \\ \text{---} \end{array} + \begin{array}{c} \text{---} \\ | \quad | \\ \text{---} \end{array} \leq 100$ Capacity of wood

$\begin{array}{c} \text{---} \\ | \quad | \\ \text{---} \end{array} + \begin{array}{c} \text{---} \\ | \quad | \\ \text{---} \end{array} \leq 80$ Upper limit on labor

A Production Problem: Formulation



Step 3: Add constraints

$$\text{subject to } 2 \text{  + \text{  } \leq 100$$

Capacity of wood

$$\text{ + \text{  } \leq 80$$

Upper limit on labor

$$\text{ } \leq 40$$

Demand for tables

A Production Problem: Formulation



Step 3: Add constraints

$$\text{subject to } 2 \text{  + \text{  \leq 100$$

Capacity of wood

$$\text{ + \text{  \leq 80$$

Upper limit on labor

$$\text{ \leq 40$$

Demand for tables

$$\text{ , \text{  \geq 0$$

Non-negativity

A Production Problem: Algebraic Formulation



Step 1: Define decision variables

Step 2: Form objective function

Step 3: Add Constraints

01

x_1 : number of tables to produce in a week
 x_2 : number of chairs to produce in a week

02

$\text{maximize}_{x_1, x_2} 3x_1 + 2x_2$ Profit

03

subject to $2x_1 + x_2 \leq 100$ Capacity of wood
 $x_1 + x_2 \leq 80$ Upper limit on labor
 $x_1 \leq 40$ Demand for tables
 $x_1, x_2 \geq 0$ Non-negativity

A Production Problem: Algebraic Formulation

Step 1:

Define decision variables

x_1 : number of
tables to produce
in a week

x_2 : number of
chairs to produce
in a week

Step 2:

Form objective function

maximize x_1, x_2
profit: $3x_1 + 2x_2$

Step 3:

Add constraints

Capacity of woods:
subject to $2x_1 + x_2 \leq 100$

Upper limit on labor
 $x_1 + x_2 \leq 80$

Demand for tables
 $x_1 \leq 40$

Non-negativity
 $x_1, x_2 \geq 0$

Optimization Terminology

Topics

- Decision Variables
 - Continuous, Discrete Variables
- Objective Function and Constraints
- Feasible/infeasible problem
- Bounded/unbounded feasible region
- Convexity

Decision Variables

- The variables that we optimize over are the **decision variables**.
- They may be **continuous variables**, meaning that they can take on values in the real numbers, including fractions, or they may be **discrete variables** that can only take on integer values.

Decision Variables

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Continuous	Discrete
Amount of raw material to mix	Number of trucks to dispatch
Investment amount in dollars	Number of workers to schedule

Objective Function and Constraints

- The function being optimized is the **objective function**.

$$\text{maximize}_{x_1, x_2} 3x_1 + 2x_2$$

- The domain over which this function is optimized is the set of points that satisfy the constraints, which can be equations or inequalities.

Objective Function and Constraints

- We distinguish two kinds of constraints:
 - **Functional constraints** can have any form,
 - **Variable bounds** restrict the allowable *values of one variable*.

$$2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

$$x_1 \leq 40$$

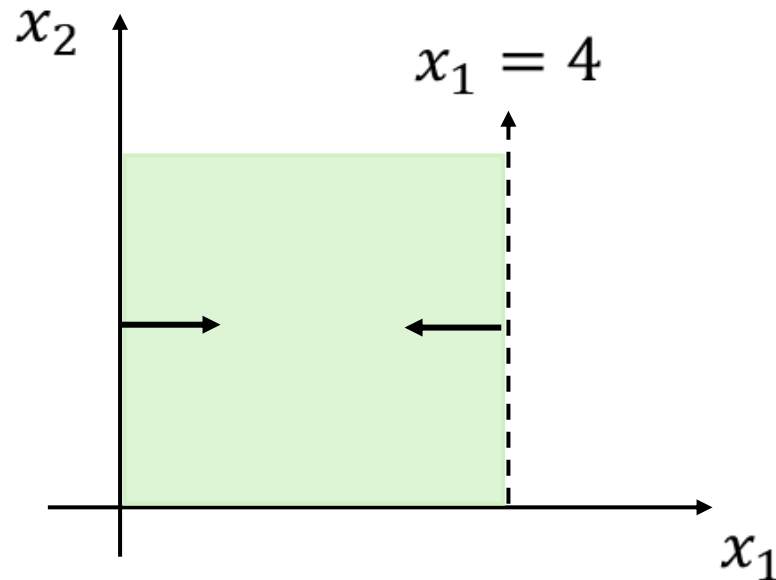
$$x_1, x_2 \geq 0$$

Solutions to Mathematical Programs

- A **solution** to a mathematical program is any value $\mathbf{x}$ of the variables.

Consider

$$\max_{x_1} \{2x_1 \in R \mid x_1 < 4, x_1 \geq 0, x_2 \geq 0\}$$



A solution to this could be:

$$x_1 = 5$$

OR

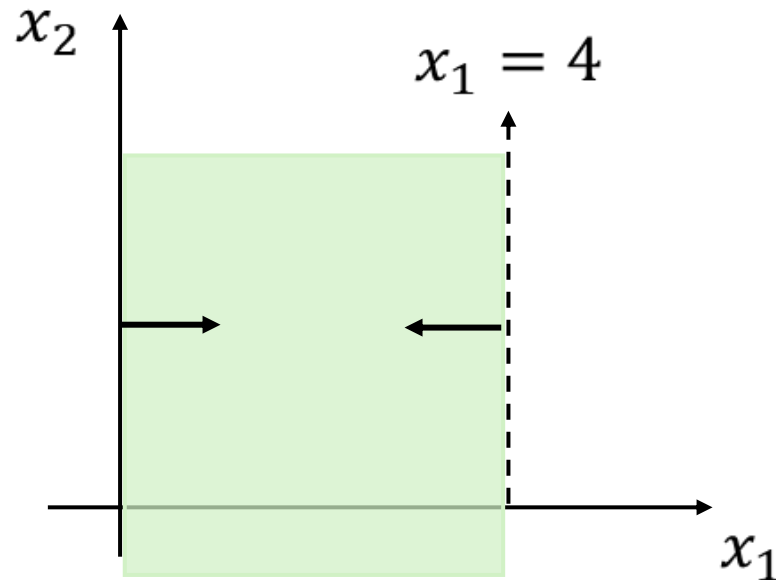
$$x_1 = -18,439$$

Feasible Solutions

- A **feasible solution** is a solution that satisfies all constraints, i.e. $\mathbf{x} \in S$.

Consider

$$\max_{x_1} \{2x_1 \in R \mid x_1 < 4, x_1 \geq 0, x_2 \geq 0\}$$



A feasible solution to this could be:

$$x_1 = 3$$

OR

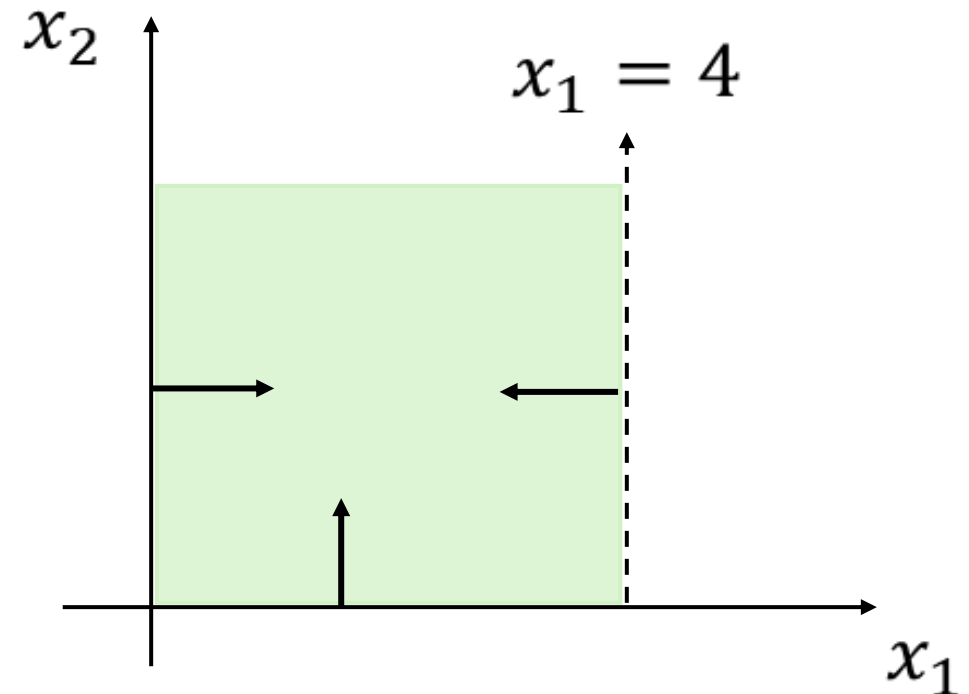
$$x_1 = 1$$

Feasible Region

- We call set of all possible solutions or points that satisfy a given set of constraints the **feasible region**

Consider

$$\max_{x_1} \{2x_1 \in R \mid x_1 < 4, x_1 \geq 0, x_2 \geq 0\}$$

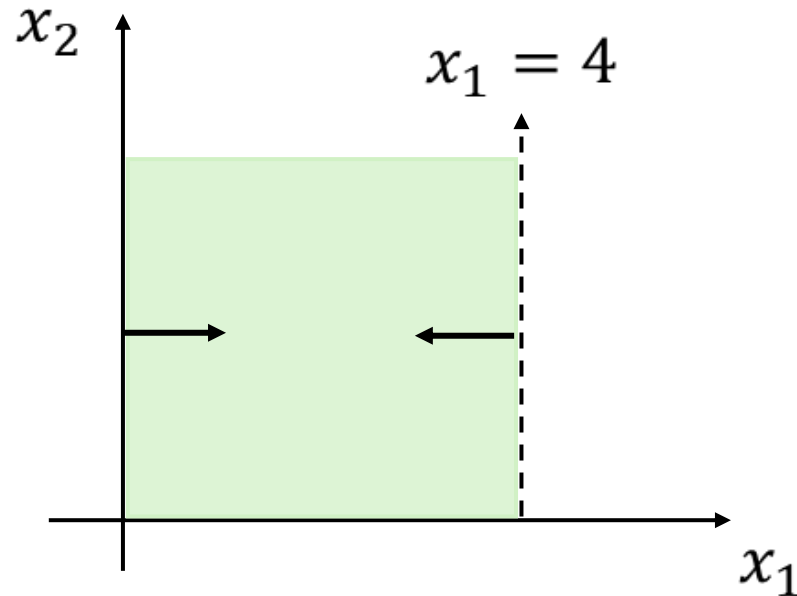


The Value of Solutions

- The **value** of a solution is its objective function value.

Consider

$$\max_{x_1} \{2x_1 \in R \mid x_1 < 4, x_1 \geq 0, x_2 \geq 0\}$$



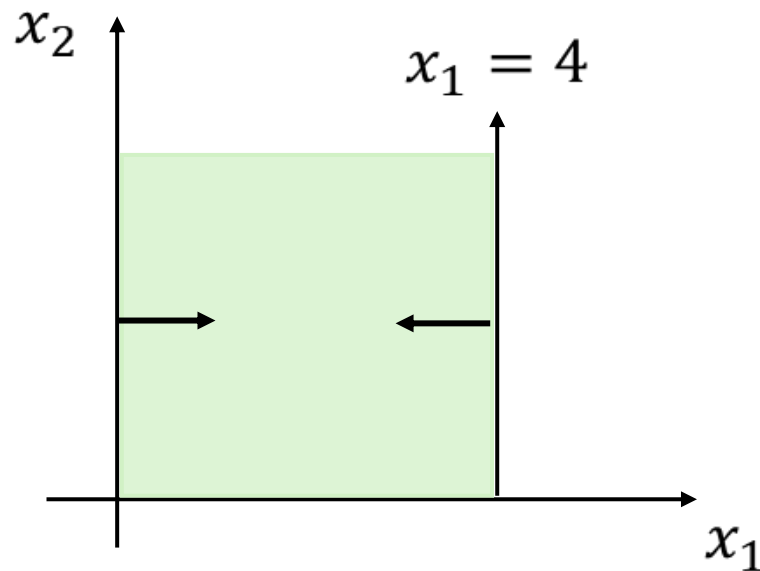
When $x_1 = 3$,
the value of a solution would be 6
And when $x_1 = 1$,
the value of a solution would be 2

The Optimal Solution

- An **optimal solution** is a feasible solution that has a value at least as large (if maximizing) as any other feasible solution.

Consider an optimization problem

$$\max_{x_1} \{2x_1 \in R \mid x_1 \leq 4, x_1 \geq 0, x_2 > 0\}$$



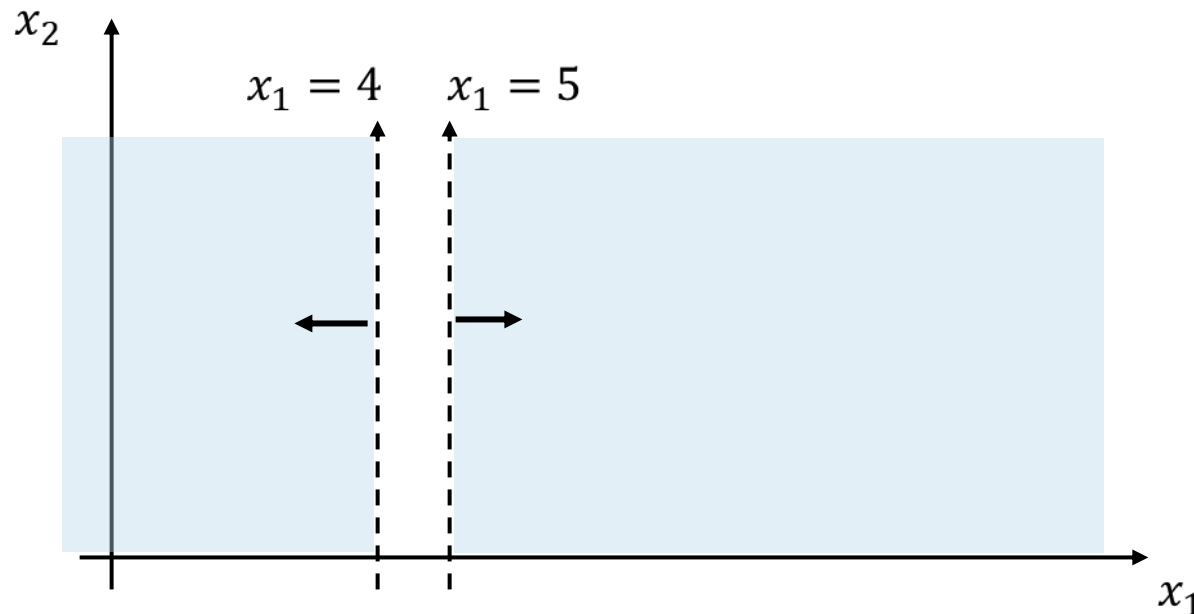
The optimal solution to this problem would be $x_1 = 4$

Existence of Optimality: Feasible vs. Infeasible Solutions

- When at least one point or solution satisfies all the constraints in a problem, that problem has a **feasible solution**

Existence of Optimality: Feasible vs. Infeasible Solutions

- When at least one point or solution satisfies all the constraints in a problem, that problem has a **feasible solution**
- Consider $\{x_1 \geq 5, x_1 \leq 4\}$. How does the feasible region look?



Infeasible problem

There are no solutions to this problem!

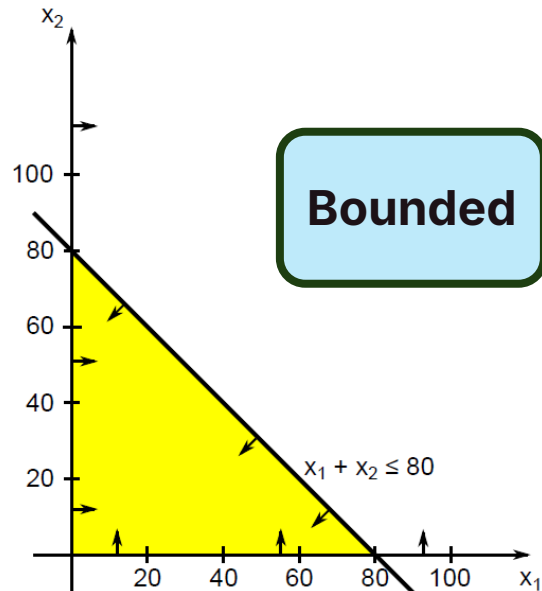
Bounded vs. Unbounded Feasible Regions

- There are two different types of feasible regions:

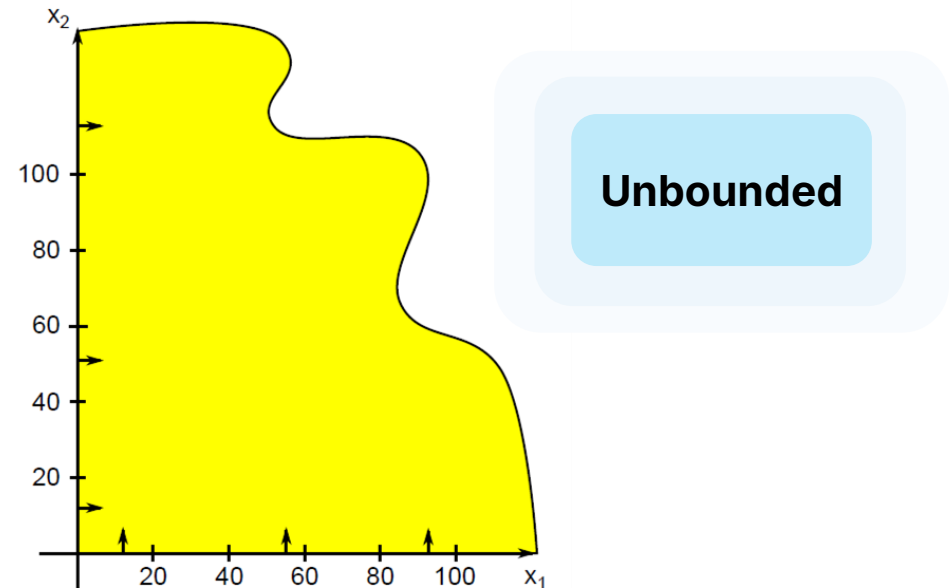
- bounded and unbounded

- Consider:

- $\{x_1 + x_2 \leq 80, x_1 \geq 0, x_2 \geq 0\}$



- $\{x_1 \geq 0, x_2 \geq 0\}$

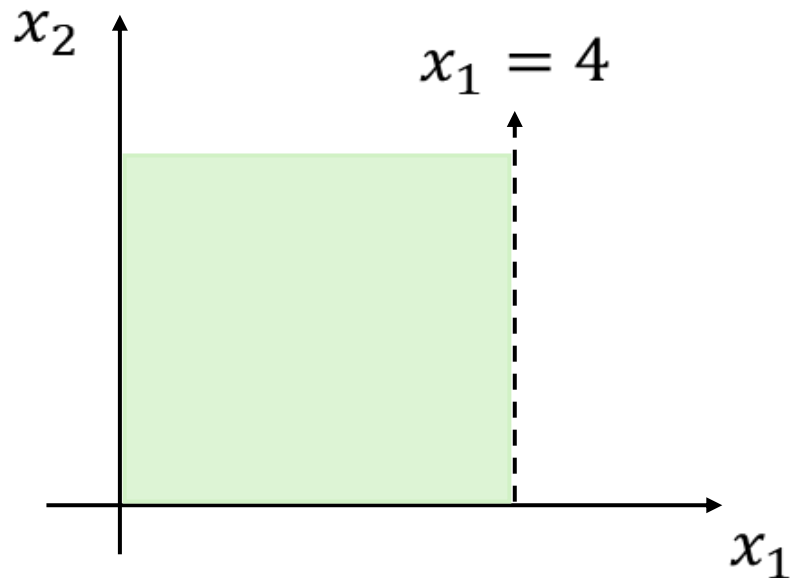


Unbounded Problem

- The problem is an **unbounded problem**, if for a maximization, there is a feasible solution $\mathbf{x}$ with value $f(\mathbf{x}) > k$ for every k

Consider an optimization problem

$$\max_{x_2} \{2x_2 \in R \mid x_1 < 4, x_1 \geq 0, x_2 \geq 0\}$$



When $k=0$, $f(k) = 0$

When $k=1$, $f(k) = 2$

When $k=4$, $f(k) = 8$,

When $k=1,000,000,000$ $f(k) = 2,000,000,000$,

...

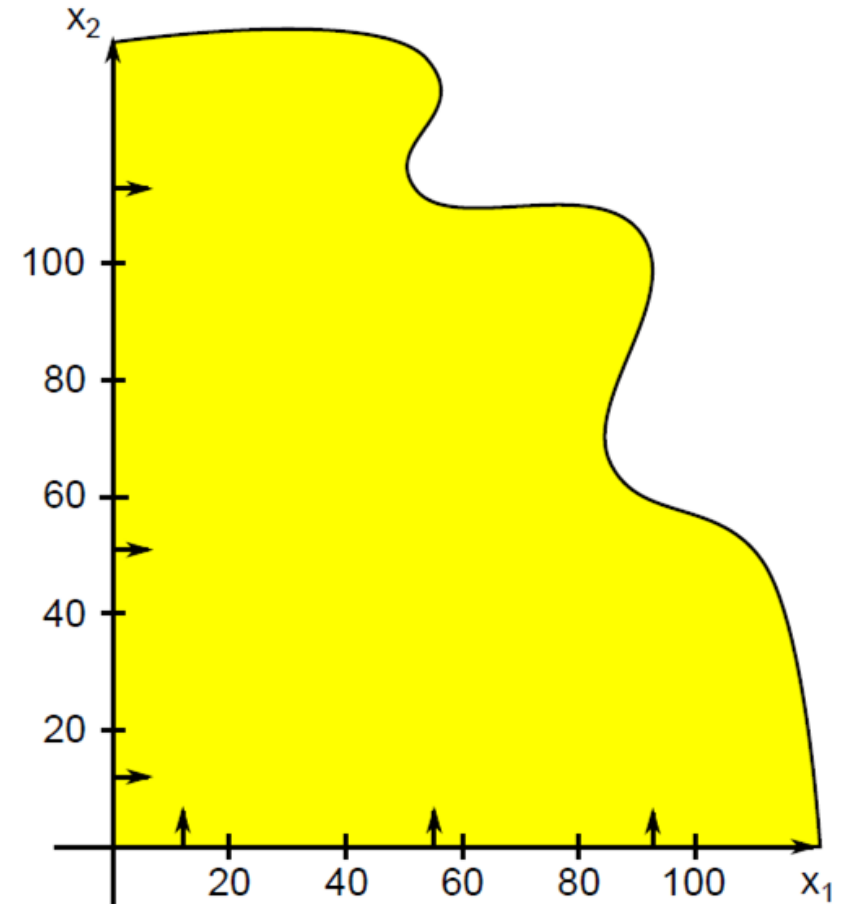
Unbounded problem

Think about:

A problem with an unbounded feasible region CAN still have an optimal solution for certain objective functions.

→ How about

$$\min_{\{x_1, x_2\}} \{x_1 + x_2 \mid x_1 \geq 0, x_2 \geq 0\}$$



Classes of Mathematical Programs

Topics

- Linear Program
- Integer Program
- Mixed Integer Program
- Binary Program
- Nonlinear Program

Linear Programming

- A **linear programming (LP)** problem is an optimization problem where:
 - We try to maximize (or minimize) a **linear function** of **continuous** decision variables.
 - Decision variables are allowed to take fractional values

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 - We try to maximize (or minimize) a **linear function** of **continuous** decision variables.
 - Decision variables are allowed to take fractional values
- The values of the decision variables must satisfy a **set of constraints**.
 - Each constraint must be a **linear equation** or **linear inequality**.

Linear Programming



Considering the definition of a linear program,
do you think the **production problem** is a linear program?

$$\begin{aligned} & \text{maximize}_{x_1, x_2} && 3x_1 + 2x_2 \\ & \text{subject to} && 2x_1 + x_2 \leq 100 \\ & && x_1 + x_2 \leq 80 \\ & && x_1 \leq 40 \\ & && x_1, x_2 \geq 0 \end{aligned}$$

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✓ **Yes!**

Both the objective
and constraints are
linear

Integer Programming

Many practical optimization problems involve making **discrete quantitative** choices

Integer Programming

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 - how many delivery trucks a logistics company should deploy to different cities



Integer Programming

- Many practical optimization problems involve making **discrete quantitative** choices, such as:
 - how many delivery trucks a logistics company should deploy to different cities
 - how many machines to purchase and install in a factory to meet production demand



Integer Programming

- Or **Logical Choices**

- whether or not each drug being developed by a pharmaceutical company should be chosen for a clinical trial



→ Both types of situations can be modeled by **integer variables**.

These variables can only take whole numbers!

Integer Programming



What if I was to say we cannot produce tables and chairs in fragments?
(e.g., we cannot produce 4.2 chairs nor 5.4 tables)

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Then the decision variables only can take integer values, and this will be an integer program

Integer Programming

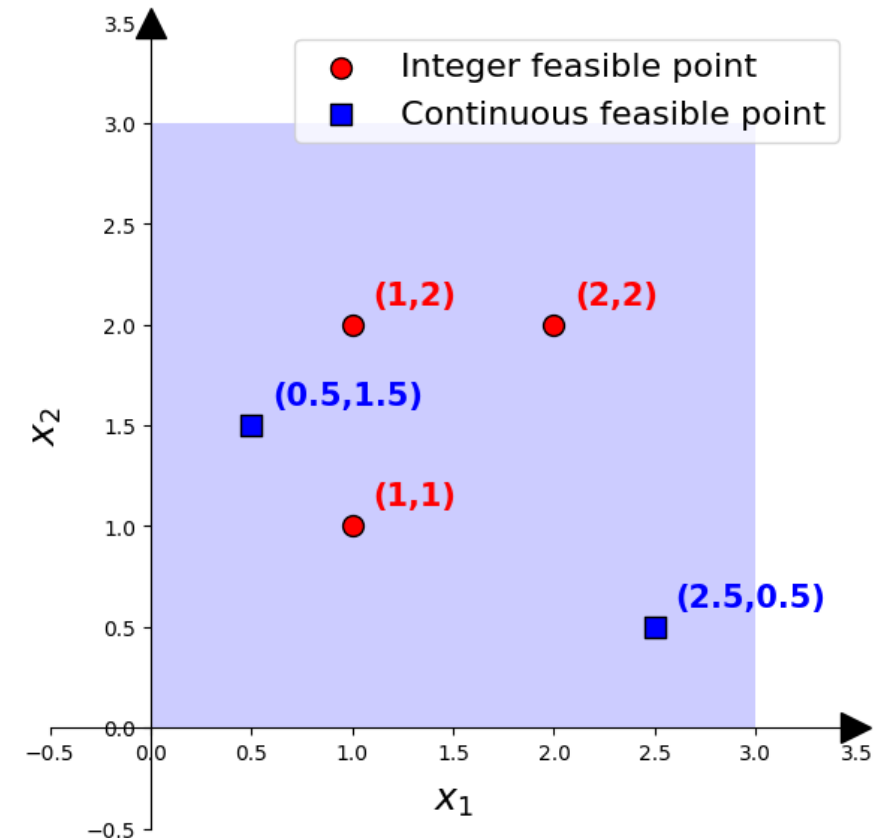
- An **integer programming (IP)** problem is an optimization problem where:
 - We try to maximize (or minimize) a **linear function** of **integer** decision variables.
 - And the decision variables **are NOT** allowed to take fractional values

Integer Programming

- An **integer programming (IP)** problem is an optimization problem where:
 - We try to maximize (or minimize) a **linear function** of **integer** decision variables.
 - And the decision variables **are NOT** allowed to take fractional values
 - The values of the decision variables must satisfy a **set of linear constraints**.

Mixed Integer Programs

- The general form of an **integer program** is the same as a **linear program** with the added constraint that the variables are integers.
- If only some of the variables are restricted to be integer, it is called a ***mixed integer program***.



Mixed Integer Programming



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Mixed Integer Programming



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Then this will
become a MILP

Binary Programming

When variables represent *logical choices*, they are usually defined so that **$x = 1$ if true** (decide to take the action) and **$x = 0$ if false** (decide not to take the action).

Binary Programming

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Binary Programming

When variables represent logical choices, they are usually defined so that **$x = 1$ if true** (decide to take the action) and **$x = 0$ if false** (decide not to take the action).

- A variable whose possible values are 0 and 1 is called a **binary variable**.
- An integer program with all binary variables is called a **binary program**.

What if my decision variables are whether to produce tables or chairs?

$$\begin{aligned} & \text{maximize}_{x_1, x_2} && 3x_1 + 2x_2 \\ & \text{subject to} && 2x_1 + x_2 \leq 100 \\ & && x_1 + x_2 \leq 80 \\ & && x_1 \leq 40 \\ & && x_1, x_2 \in \{0, 1\} \end{aligned}$$

Binary Programming



What if my decision variables are whether to produce tables or chairs?

$$\text{maximize}_{x_1, x_2} \quad 3x_1 + 2x_2$$

$$\text{subject to} \quad 2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

$$x_1 \leq 40$$

$$x_1, x_2 \in \{0, 1\}$$



Then this will
become a BP

Nonlinear Programming

- **Nonlinear program (NLP):** An optimization problem where the objective function and constraints may be nonlinear

Nonlinear Programming

- **Nonlinear program (NLP):** An optimization problem where the objective function and constraints may be nonlinear
- The variables are assumed to be continuous, as in a linear program.

Nonlinear Programming



What if the profit made from selling a table was the square of the number of tables produced?

$$\begin{aligned} & \text{maximize}_{x_1, x_2} \quad 3x_1^2 + 2x_2 \\ & \text{subject to} \quad 2x_1 + x_2 \leq 100 \\ & \quad \quad \quad x_1 + x_2 \leq 80 \\ & \quad \quad \quad x_1 \leq 40 \\ & \quad \quad \quad x_1, x_2 \in \{0, 1\} \end{aligned}$$

Nonlinear Programming



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Then this will
become an NLP

In a Nutshell

Program Type	Decision Variables	Objective and Constraints
Linear Program	Continuous	Linear functions
Integer Program	Integer	Linear functions
Mixed Integer Linear Program	Both integer and continuous	Linear functions
Binary Program	Binary	Linear functions
Nonlinear Program	Continuous	Nonlinear functions

Example Problem

A Disaster Aid Problem

A Disaster Aid Problem



After a major disaster, an aid organization wants to quickly send tents and food using a cargo plane.

Maximize the combined benefit from tents and food sent:

- The total weight of **all pallets** cannot exceed **40,000 lbs**
- There is space for at most **6 pallets** on a plane
- Only up to **5 pallets of food** are currently available to send
- We **cannot send a negative number** of pallets
- The benefits are **8 units per each pallet of tent** and **6 units per each pallet of food** sent
- The following table lists the weight, cost, criticality, expected benefit of each pallet.

A Disaster Aid Problem



Table: Data for Sending Aid

	Tents	Food
Weight (1000 lbs)	7.5	5
Cost (\$1000/pallet)	11	4
Criticality	5	8
Expected Benefit	8	6

**How many pallets of tents and food must be sent
to achieve the highest possible benefit?**

A Disaster Aid Problem: Formulation



Step 1: Define decision variables

01



How many pallets of tent should we send in a cargo plane?



How many pallets of food should we send in a cargo plane?

A Disaster Aid Problem: Formulation



Step 2: Form objective function

What is our aim/objective?

To find the loading plan that will **maximize** the benefit

$$\text{Benefit: } 8 \text{  + 6 \text{ $$

02

$$\text{maximize } 8 \text{  + 6 \text{ $$

A Disaster Aid Problem: Formulation



Step 3: Add constraints

x



produced each week

y



produced each week

03

subject to

$7.5x + 5y \leq 40$	<i>Weight capacity</i>
$x + y \leq 6$	<i>Space limit</i>
$y \leq 5$	<i>Food available</i>
$x, y \geq 0$	<i>Non-negativity clause</i>

A Disaster Aid Problem: Formulation



Step 4: Complete problem

max $8x + 6y$ *Objective function*

s.t $7.5x + 5y \leq 40$ *Constraints*

$$x + y \leq 6$$

$$y \leq 5$$

$$x, y \geq 0$$

Discussion: Integer vs Linear Programs

- In **integer programs**, the optimal solution is not necessarily obtained by rounding the **linear program**.
 - In fact, there is no limit to how far the optimal solution could be from the linear programming solution.
- The **integer program** can be infeasible when the **linear program** is feasible.
- These difficulties suggest that **integer programs** are harder to solve than **linear programs**, which we will see is true.